

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
Student Name:	Leo Leenas A	Reg. No.:	192511068
Date:	10/10/2026	Duration:	60 minutes

Code of Conduct

I Leo Leenas A (192511068) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

- 1. Identify the Project Vision and Stakeholders**
 - Define the project's vision, objectives, and identify all key stakeholders involved.
- 2. Create Epics and User Stories**
 - Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.
- 3. Define Acceptance Criteria**
 - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
- 4. Prioritize the Product Backlog**
 - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
- 5. Plan the Sprint Backlog**
 - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
- 6. Estimate Effort Using Story Points**
 - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
- 7. Present the Sprint Goal and Expected Deliverables**
 - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
Acceptance Criteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5
TOTAL	/25

Question Number: 18

Online Banking Service Management System

Problem Statement: Develop a secure platform for customers to view accounts, transfer funds, pay bills, download statements, and receive transaction notifications.

Approach: This document is prepared as an Agile Sprint Planning and User Story Workshop using the Scrum framework. The workshop is carried out in the same order as the student instructions: first the vision and stakeholders are agreed, then the requirements are split into epics and written as user stories, acceptance criteria are added, the product backlog is prioritised, the stories for Sprint 1 are selected and broken into tasks, each story is estimated with Planning Poker, and finally the sprint goal and deliverables are fixed. Figure 1 shows the Scrum cycle that this workshop belongs to.

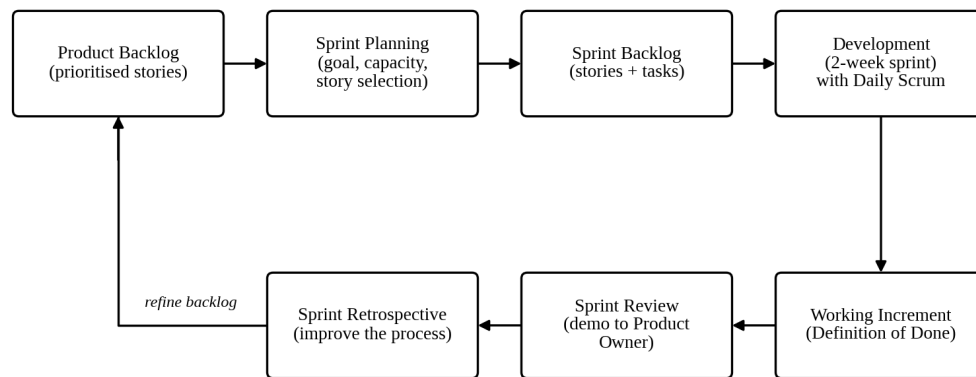


Figure 1: Scrum cycle followed in the workshop

1. Project Vision and Stakeholders

1.1 Project Vision

Vision statement: For bank customers who want to manage their money without visiting a branch, the Online Banking Service Management System is a secure web-based banking platform that lets them view accounts, transfer funds, pay bills, download statements and receive instant transaction alerts. Unlike manual branch banking, it is available 24 x 7 and protects every login and transaction with OTP verification.

1.2 Project Objectives

- Provide secure customer login with two-factor (OTP) verification for every user.
- Show account balances and recent transactions within 3 seconds of login.
- Allow a customer to complete a fund transfer in three simple steps, in under one minute.
- Allow payment of utility bills to registered billers with an instant receipt.
- Generate downloadable PDF statements for any period up to 12 months within 10 seconds.

- Send an SMS and e-mail alert within one minute of every debit or credit.
- Give administrators full control and audit visibility, in line with banking compliance rules.

1.3 Key Stakeholders

Everyone who uses, builds, approves or is affected by the system is listed below. Figure 2 shows how they connect to the product.

- **Bank customers (primary users):** use the system to check accounts, transfer money, pay bills and get statements. They need speed, safety and ease of use.
- **Product Owner:** owns the product backlog, decides priorities and accepts completed stories.
- **Scrum Master:** facilitates the sprint events, removes blockers and makes sure Scrum practices are followed.
- **Development team:** developers, a UI designer and a database engineer who design, code and integrate the features.
- **QA / test team:** verify every story against its acceptance criteria and report defects.
- **Bank administrators:** manage customer accounts, monitor transactions and view audit logs.
- **Customer support team:** handle customer queries and failed-transaction complaints.
- **Compliance and security officer:** makes sure KYC, data security and regulatory (RBI) guidelines are met.
- **SMS, e-mail and payment providers:** external partners who supply the gateways for OTP, notifications, fund transfer and bill payment.
- **Bank management (sponsor):** funds the project and expects fewer branch visits and better customer satisfaction.

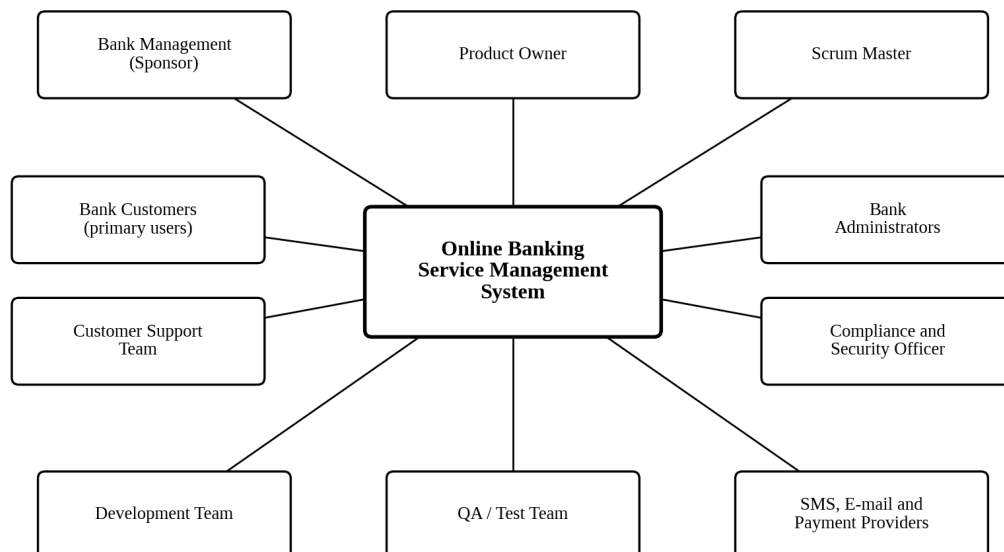


Figure 2: Stakeholder map of the Online Banking Service Management System

2. Epics and User Stories

2.1 Epics

In the user story workshop the requirements in the problem statement were first grouped into seven epics. An epic is a large piece of work that is too big for one sprint and is therefore split into smaller user stories. Figure 3 shows the epics and the stories that belong to each of them.

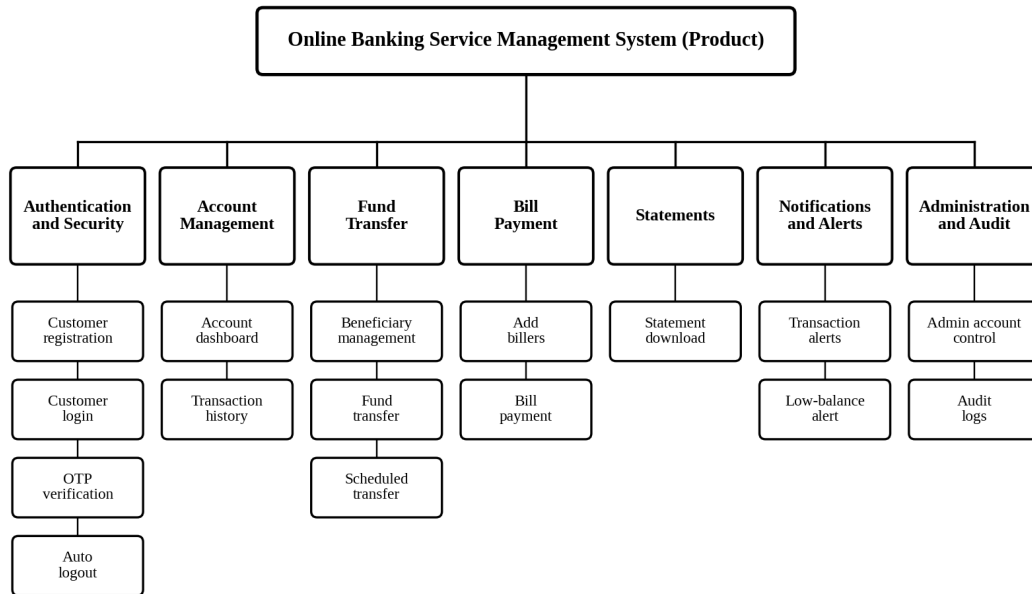


Figure 3: Epics and their user stories

2.2 User Stories

Every story is written in the format *As a <user>, I want <feature>, so that <benefit>*. The stories were checked against the INVEST rules (Independent, Negotiable, Valuable, Estimable, Small and Testable) so that each one can be finished inside a single sprint.

Epic 1 - Authentication and Security

Registration, login, OTP verification and session protection.

- **Customer registration:** As a new customer, I want to register for internet banking using my account number and registered mobile number, so that I can access banking services online.
- **Customer login:** As a registered customer, I want to log in with my user ID and password, so that only I can access my accounts.
- **OTP verification:** As a customer, I want to verify my login and every transaction with a one-time password (OTP), so that my account stays safe from unauthorised access.
- **Auto logout:** As a customer, I want to be logged out automatically after 5 minutes of inactivity, so that my account is protected if I leave my device unattended.

Epic 2 - Account Management

Viewing accounts, balances and transaction history.

- **Account dashboard:** As a customer, I want to see all my accounts with their current balances on a dashboard, so that I know my financial position at a glance.
- **Transaction history:** As a customer, I want to view my recent transactions and filter them by date or type, so that I can keep track of my spending.

Epic 3 - Fund Transfer

Beneficiary management and transfer of money.

- **Beneficiary management:** As a customer, I want to add and manage beneficiaries, so that I can send money to them quickly in future.
- **Fund transfer:** As a customer, I want to transfer funds to my beneficiary's account, so that I can send money without visiting the branch.
- **Scheduled transfer:** As a customer, I want to schedule a transfer for a future date, so that I can make regular payments such as rent on time.

Epic 4 - Bill Payment

Adding billers and paying utility bills.

- **Add billers:** As a customer, I want to add billers such as electricity, water and mobile, so that I can manage all my bills in one place.
- **Bill payment:** As a customer, I want to pay a bill from my account and receive a payment receipt, so that I have proof of payment.

Epic 5 - Statements

Generating and downloading account statements.

- **Statement download:** As a customer, I want to download my account statement as a PDF for a chosen period, so that I can use it for loan or tax purposes.

Epic 6 - Notifications and Alerts

SMS and e-mail alerts for transactions and low balance.

- **Transaction alerts:** As a customer, I want to receive an SMS and e-mail alert for every debit and credit, so that I can quickly notice any suspicious activity.
- **Low-balance alert:** As a customer, I want to set a low-balance alert, so that I can avoid failed payments due to insufficient funds.

Epic 7 - Administration and Audit

Account control by administrators and audit trail for compliance.

- **Admin account control:** As a bank administrator, I want to view, block or unblock customer accounts, so that I can respond quickly to fraud or customer requests.
- **Audit logs:** As a bank administrator, I want to view audit logs of all logins and transactions, so that I can ensure compliance and investigate issues.

3. Acceptance Criteria

Acceptance criteria are written in the Given-When-Then format for the seven user stories selected for Sprint 1. Each criterion is specific and can be tested directly by the QA team.

Customer registration

- **Criterion 1:** **Given** a customer has an active account and a registered mobile number, **when** he enters the correct account number, mobile number and date of birth, **then** an OTP is sent to his mobile and, once verified, he is allowed to create a user ID and password.
- **Criterion 2:** **Given** the entered details do not match the bank's records, **when** he submits the registration form, **then** registration is rejected with a general message and the system does not reveal which field was wrong.
- **Criterion 3:** **Given** the customer is creating a password, **when** the password has fewer than 8 characters or lacks an upper-case letter, a number or a special character, **then** the password is rejected and the password rules are displayed.

Customer login

- **Criterion 1:** **Given** a registered customer is on the login page, **when** he enters the correct user ID and password, **then** the system accepts the credentials and moves to the OTP verification step.
- **Criterion 2:** **Given** the customer enters a wrong password, **when** he fails three times in a row, **then** the account is locked for 30 minutes and a message is shown.

OTP verification

- **Criterion 1:** **Given** valid credentials have been entered, **when** a 6-digit OTP is sent and the customer enters it correctly within 3 minutes, **then** access to the dashboard is granted.
- **Criterion 2:** **Given** the OTP is wrong or has expired, **when** the customer submits it, **then** access is denied, an error is shown and a resend option is offered.
- **Criterion 3:** **Given** an OTP has already been used once, **when** the same OTP is entered again, **then** it is rejected, because each OTP can be used only once.

Account dashboard

- **Criterion 1:** **Given** the customer is logged in and holds one or more accounts, **when** the dashboard loads, **then** every account is listed with a masked account number (last 4 digits only) and its available balance within 3 seconds.
- **Criterion 2:** **Given** a transfer has just been completed, **when** the customer refreshes the dashboard, **then** the balance shows the updated amount.

Beneficiary management

- **Criterion 1:** **Given** the customer is logged in, **when** he enters the beneficiary name, account number and IFSC code and confirms with an OTP, **then** the beneficiary is saved and appears in the beneficiary list.
- **Criterion 2:** **Given** the IFSC code is invalid or the account number is already in the list, **when** he tries to save the beneficiary, **then** the system shows an error and does not save the entry.
- **Criterion 3:** **Given** a beneficiary exists in the list, **when** the customer chooses to delete it and confirms, **then** the beneficiary is removed from the list.

Fund transfer

- **Criterion 1:** **Given** the balance is sufficient and a valid beneficiary is selected, **when** the customer enters the amount and confirms with an OTP, **then** the amount is debited, credited to the beneficiary and a unique reference number is displayed.
- **Criterion 2:** **Given** the balance is less than the amount entered, **when** the customer submits the transfer, **then** the transfer is blocked with the message "Insufficient balance" and nothing is debited.
- **Criterion 3:** **Given** the amount is above the daily limit of Rs. 1,00,000, **when** the customer submits the transfer, **then** the transfer is blocked and the limit is shown to the customer.
- **Criterion 4:** **Given** a technical failure occurs during the transfer, **when** the transaction is processing, **then** the whole transaction is rolled back so that no partial debit remains.

Transaction alerts

- **Criterion 1:** **Given** a debit or credit is posted successfully, **when** the transaction is completed, **then** an SMS and an e-mail are sent within 1 minute showing amount, type, date and time, masked account number and available balance.
- **Criterion 2:** **Given** a transaction fails, **when** the failure is recorded, **then** no debit or credit alert is sent and the customer sees a failure message on screen.

4. Product Backlog Prioritization

The backlog is ordered using three factors: **business value** (how much the bank and the customer gain), **customer need** (how often the feature will be used) and **dependency** (what must be built first). Priority is given with the MoSCoW method. Security and the core money-movement features come first because nothing else can be released safely without them. The number in brackets is the story point estimate.

Must Have

- **1. Customer registration (5):** no dependency, and every other feature needs a registered user.
- **2. Customer login (3):** depends on customer registration; gives access to everything else.
- **3. OTP verification (5):** depends on customer login; protects login and every sensitive action.
- **4. Account dashboard (3):** depends on customer login; the first thing customers want to see.
- **5. Beneficiary management (5):** depends on OTP verification; needed before any transfer.
- **6. Fund transfer (8):** depends on OTP verification, account dashboard and beneficiary management; the highest-value feature of the product.
- **7. Transaction alerts (3):** depends on fund transfer; every transfer must notify the customer from the first release.
- **8. Transaction history (3):** depends on account dashboard; customers need it to verify their transactions.

Should Have

- **9. Auto logout (3):** depends on customer login; improves safety on shared devices.
- **10. Add billers (5):** depends on OTP verification.
- **11. Bill payment (5):** depends on account dashboard and add billers.
- **12. Statement download (5):** depends on transaction history.
- **13. Admin account control (5):** depends on customer login.
- **14. Audit logs (5):** depends on fund transfer; needed for compliance.

Could Have

- **15. Low-balance alert (2):** depends on transaction alerts; useful but not essential.
- **16. Scheduled transfer (8):** depends on fund transfer; can wait until the basic transfer is stable.

Figure 4 shows the dependency order of the stories picked for Sprint 1 and the stories that remain in the backlog.

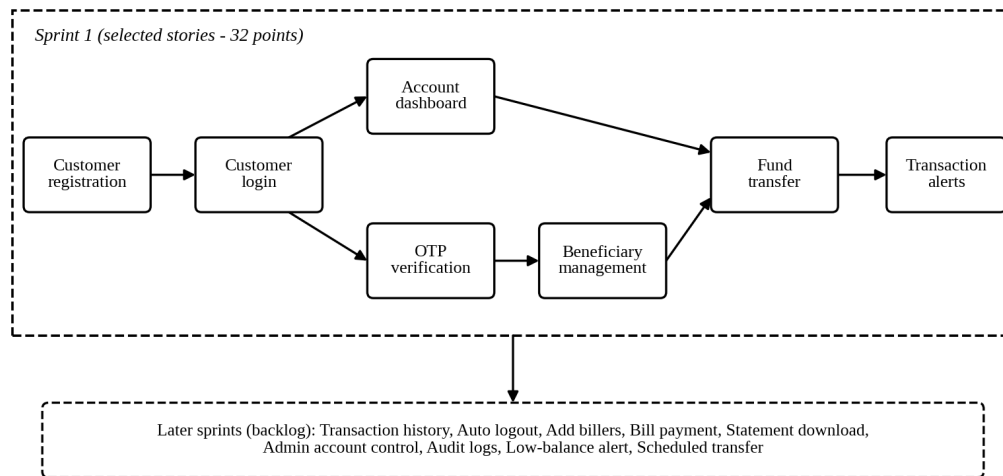


Figure 4: Dependency order of the Sprint 1 stories

5. Sprint Backlog (Sprint 1)

Sprint length: 2 weeks (10 working days). **Team:** 5 members. **Available capacity:** about 300 working hours (5 members x 10 days x 6 productive hours). The seven highest-priority stories are selected for Sprint 1, which comes to 32 story points. Each story is divided into development tasks, with the planned hours given in brackets.

- **Customer registration (5 points, 30 hours):** design the registration form UI (6 h); create the customer and credential tables (4 h); build the registration API with field validation (10 h); hook OTP verification into registration (4 h); write unit tests (6 h).
- **Customer login (3 points, 18 hours):** build the login page (4 h); add password hashing (bcrypt) and a session token (6 h); add account lock after 3 failed attempts (4 h); write unit tests (4 h).
- **OTP verification (5 points, 32 hours):** integrate the SMS gateway (sandbox) (8 h); generate and store a hashed 6-digit OTP with 3-minute expiry (8 h); build the OTP entry screen (6 h); add resend and attempt-limit logic (4 h); write unit tests (6 h).
- **Account dashboard (3 points, 19 hours):** design the dashboard UI (6 h); build the account summary API (6 h); mask account numbers (2 h); check integration and response time (5 h).
- **Beneficiary management (5 points, 31 hours):** create the beneficiary table and API (8 h); build the add-beneficiary form with IFSC validation (8 h); add the OTP confirmation step (4 h); build list and delete options (5 h); write unit tests (6 h).
- **Fund transfer (8 points, 60 hours):** build the transfer API using a database transaction (14 h); add balance and daily-limit validation (8 h); build the three-step transfer screen (12 h); generate the reference number and receipt page (6 h); handle rollback on failure (8 h); run integration testing (12 h).
- **Transaction alerts (3 points, 21 hours):** build an event-based notification service (8 h); prepare SMS and e-mail templates (4 h); integrate the SMS/e-mail sandbox (5 h); test delivery time (4 h).

The total planned effort is 211 hours, which is about 70% of the available capacity. The remaining time is kept as a buffer for sprint meetings, code reviews and bug fixing. The stories not selected stay in the product backlog for later sprints.

6. Effort Estimation Using Story Points

Technique: Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13). Each team member privately picks a card, the cards are shown together, and any large difference is discussed before voting again until the team agrees. The **account dashboard** was taken as the reference story and fixed at 3 points because it is a simple, read-only screen. Every other story was compared against it.

- **Customer registration - 5 points:** needs form validation, matching against bank records, password rules and a link to OTP verification.
- **Customer login - 3 points:** standard login with password hashing and a lock rule; similar in size to the reference story.
- **OTP verification - 5 points:** depends on an external SMS gateway and needs expiry, single use and retry limits, so the risk is moderate.
- **Account dashboard - 3 points:** the reference story; a read-only screen using existing account data.
- **Beneficiary management - 5 points:** add, list and delete operations with IFSC validation and an OTP confirmation step.
- **Fund transfer - 8 points:** the most complex story. It involves money movement, database transactions, limit checks, rollback and several screens, so it carries the highest risk.
- **Transaction alerts - 3 points:** event-based notification with templates; it depends on the SMS and e-mail sandbox but the logic is simple.

The total for Sprint 1 is 32 story points. As this is the first sprint there is no past velocity, so a velocity of 30 to 35 points is assumed for a team of five and a commitment of 32 points is realistic. The velocity will be corrected after the sprint review.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

"By the end of Sprint 1, a registered customer will be able to log in securely using an OTP, view the account balance, add a beneficiary, transfer money to that beneficiary and receive an alert for the transaction."

7.2 Expected Deliverables

- Customer registration and login screens with password rules and account lock.
- OTP generation and verification for login and for sensitive actions.
- Dashboard showing all accounts with masked numbers and current balances.
- Beneficiary add, list and delete function with IFSC validation.
- Working fund transfer with balance check, daily limit, rollback and reference number.
- SMS and e-mail alerts for each successful debit and credit.
- A tested, demo-ready increment with unit-test and integration-test results.

Figure 5 shows the parts of the system that will be built and connected by the end of the sprint.

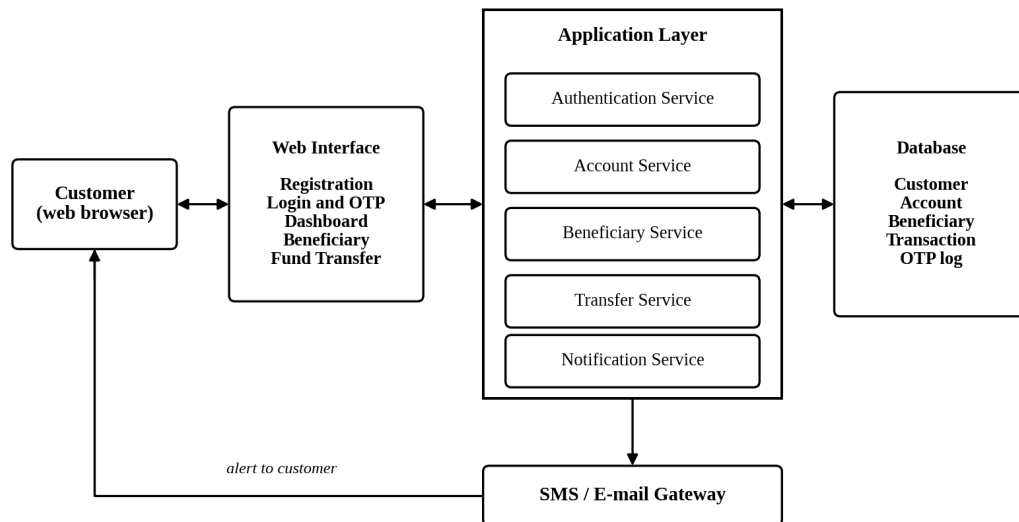


Figure 5: Block diagram of the system delivered in Sprint 1

7.3 Definition of Done

- Code is completed, reviewed and merged to the main branch.
- All acceptance criteria of the story are tested and passed.
- No open high-priority defects remain.
- Passwords and OTPs are stored only in hashed form and account numbers are masked on screen.
- The Product Owner has accepted the story in the sprint review.

7.4 Sprint Schedule

- **Day 1:** sprint planning to confirm the sprint goal, stories and tasks.
- **Day 2 to Day 8:** development and testing, with a 15-minute daily stand-up.
- **Day 9:** sprint review, where the working increment is demonstrated to the Product Owner and stakeholders.
- **Day 10:** sprint retrospective to discuss what went well, what did not and what to improve in Sprint 2.

7.5 Risks and Mitigation

- **Delay in SMS gateway access for OTP:** use a sandbox gateway during development and begin the integration on Day 2.
- **Fund transfer story (8 points) overruns:** start it early in the sprint and keep the scheduled transfer story out of this sprint.
- **Security flaws in login and transfer:** peer code review and testing of every wrong-input scenario before the demo.

Conclusion: Sprint 1 delivers the core and most valuable part of the Online Banking Service Management System: secure access, account visibility and safe money transfer with notifications. Bill payment, statements and administration features follow in the next sprints according to the product backlog.